

Ex.No.7 Use AFLogical OSE to extract data from an Android device

Description

AFLogical OSE (Open Source Edition) is a tool used to extract data from Android devices. It's part of the Open Source Android Forensics project and is often used for logical extraction, which means pulling data like contacts, messages, and call logs without directly accessing the file system. Here's a step-by-step guide to using AFLogical OSE to extract data from an Android device.

Step 1: Prepare Your Environment

1. Download AFLogical OSE:

- Visit the [AFLogical OSE GitHub page](#) and download the latest release. Alternatively, you can clone the repository using Git.

2. Install Java (if not already installed):

- AFLogical OSE requires Java to run. Make sure you have Java installed on your machine. You can download it from the [official Java website](#).

3. Install Android Debug Bridge (ADB):

- ADB is a command-line tool that allows communication with Android devices. Download it from the [Android Developer's website](#) and install it on your system.
- Add ADB to your system's PATH environment variable to easily run ADB commands from the terminal or command prompt.

4. Enable USB Debugging on the Android Device:

- On the Android device, go to Settings > About Phone and tap Build Number seven times to enable Developer Options.
- Go to Settings > Developer Options and enable USB Debugging.

Step 2: Connect the Android Device to Your Computer

1. Connect via USB:

- Use a USB cable to connect the Android device to your computer.

2. Verify ADB Connection:

- Open Command Prompt (Windows) or Terminal (macOS/Linux) and run:

```
bash
```

```
adb devices
```

- This command lists connected devices. If your device is listed, you are ready to proceed. If not, ensure that USB Debugging is enabled and that the correct drivers are installed.

Step 3: Extract Data Using AFLogical OSE

1. Run AFLogical OSE:

- Navigate to the directory where you have downloaded or extracted AFLogical OSE.

2. Push the APK to the Android Device:

- Run the following command to install the AFLogical OSE APK on the device:

bash

```
adb install aflogical.apk
```

- This installs the AFLogical app on the device, which will be used to collect data.

3. Launch the AFLogical OSE App on the Device:

- On the Android device, open the AFLogical app.
- You'll see options to extract various types of data (e.g., contacts, SMS, MMS, call logs). Select the data types you want to extract.

4. Start Data Extraction:

- Once you've selected the data types, the app will start the extraction process.
- Data is stored in .csv files on the device's storage, typically in a directory named aflogical.

Step 4: Transfer Extracted Data to Your Computer

1. Use ADB to Pull Data:

- After extraction, you need to transfer the data files from the Android device to your computer. Run:

bash

```
adb pull /sdcard/aflogical /path/to/destination
```

- Replace /path/to/destination with the directory path where you want to save the extracted data on your computer.

1. Verify the Data:

- Navigate to the destination directory and check the .csv files to ensure all required data has been extracted.

Step 5: Analyze the Extracted Data

1. Open the CSV Files:

- Use Excel, Google Sheets, or any text editor to open and analyze the .csv files. These files contain the extracted data like contacts, SMS, MMS, and call logs.

2. Review and Document:

- Carefully review the data for any relevant information. Document your findings and consider exporting the data into a report format for further analysis.

Step 6: Clean Up

1. Uninstall AFLogical OSE:

- Once data extraction is complete, you can uninstall the AFLogical OSE app from the Android device by running:

bash

```
adb uninstall com.viaforensics.android.aflogical
```

2. Disconnect the Device:

- Safely disconnect the Android device from your computer.

By following these steps, you can use AFLogical OSE to effectively extract and analyze data from an Android device. This process is valuable for forensic investigations, especially when dealing with logical data that doesn't require root access to the device.

Output:

1. Installation of Java and ADB Tools

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\saiha\Downloads\platform-tools-latest-windows\platform-tools> java -version
java version "1.8.0_503"
Java(TM) SE Runtime Environment (build 1.8.0_503-b01)
Java HotSpot(TM) 64-Bit Server VM (build 25.503-b01, mixed mode)
PS C:\Users\saiha\Downloads\platform-tools-latest-windows\platform-tools> .\adb.exe version
Android Debug Bridge version 1.0.41
Version 37.0.1-15733141
Installed as C:\Users\saiha\Downloads\platform-tools-latest-windows\platform-tools\adb.exe
Running on Windows 10.0.26200
PS C:\Users\saiha\Downloads\platform-tools-latest-windows\platform-tools> |
```

2. Execution of aflogical.apk (Installation on Device)

```
PS C:\Users\saiha\Downloads\platform-tools-latest-windows\platform-tools> .\adb.exe install --bypass-low-target-sdk-block aflogical.apk
Performing Streamed Install
Success
PS C:\Users\saiha\Downloads\platform-tools-latest-windows\platform-tools> |
```

3. Extracted SMS Logs and Call Logs of Android Device

```
PS C:\Users\saiha\Downloads\platform-tools-latest-windows\platform-tools> .\adb.exe shell content query --uri content://sms > sms_extraction.txt
PS C:\Users\saiha\Downloads\platform-tools-latest-windows\platform-tools> .\adb.exe shell content query --uri content://call_log/calls > call_logs.txt
PS C:\Users\saiha\Downloads\platform-tools-latest-windows\platform-tools> dir *.txt

Directory: C:\Users\saiha\Downloads\platform-tools-latest-windows\platform-tools

Mode                LastWriteTime         Length Name
----                -
-a----             07-09-2026   19:05       12309828 call_logs.txt
-a----             07-09-2026   18:09        1154131 NOTICE.txt
-a----             07-09-2026   19:05       3311374 sms_extraction.txt

PS C:\Users\saiha\Downloads\platform-tools-latest-windows\platform-tools> |
```

4. Output of Extracted SMS Logs (sms_extraction.txt)

```
File Edit View H1 B I X S A T P while i obser sms x +  
Row: 0 _id=3359, thread_id=461, address=AA-AIRTEL-P, person=NULL, date=1788778845375, date_sent=1788778844000, protocol=0, read=0, status=-1, type=1, reply_path_present=0,  
subject=proto:cjokIMkVbSnb29nbgUwYskmcpZC5chHBZLml1c3NhZ2Zluz4SFClAKHdJfbyNEgNPYEy7AgLWmevN, body=Introducing Airtel2's Box Office Pack! Get Amazon Prime, Zee5, JioHotstar, 19 OTTs and  
30GB - all for just Rs. 200. Recharge now https://i.airtel.in/rc200s1l, creator=com.google.android.apps.messaging, seen=1  
Row: 1 _id=3358, thread_id=381, address=AD-6500022-P, person=NULL, date=1788772227597, date_sent=1788772224000, protocol=0, read=0, status=-1, type=1, reply_path_present=0,  
subject=proto:cjokIMkVbSnb29nbgUwYskmcpZC5chHBZLml1c3NhZ2Zluz4SFClAKHAi80T7MlBKr6tCYK4Ohq, body=Earn up to Rs.250 CASHBACK!  
Refer a friend to open a Safe Second Account with Airtel Payments Bank & get rewarded now - https://i.airtel.in/Get_Rewards, service_center=+919890784018, locked=0, sub_id=1,  
error_code=-1, creator=com.google.android.apps.messaging, seen=1  
Row: 2 _id=3357, thread_id=703, address=VM-MMPTR1-P, person=NULL, date=1788755449738, date_sent=1788755447000, protocol=0, read=0, status=1, type=1, reply_path_present=0,  
subject=proto:cjokIMkVbSnb29nbgUwYskmcpZC5chHBZLml1c3NhZ2Zluz4SFClAKHc3LSNF-CBhq9VmKwe9nf, body=For your Lucknow trip, you've earned up to 20% OFF* on stays with your flight booking.  
Use code: BOOKSTAYS at app.mvty.com/XmzV/Irlmkyp | Team MakeMyTrip, service_center=+919820405114, locked=0, sub_id=1, error_code=-1, creator=com.google.android.apps.messaging, seen=1  
Row: 3 _id=3356, thread_id=461, address=AA-AIRTEL-P, person=NULL, date=1788681512612, date_sent=1788681507000, protocol=0, read=0, status=1, type=1, reply_path_present=0,  
subject=proto:cjokIMkVbSnb29nbgUwYskmcpZC5chHBZLml1c3NhZ2Zluz4SFClAKHdyI57ovU9eYmJSzkTVH5As, body=Introducing Airtel2's Box Office Pack! Get Amazon Prime, Zee5, JioHotstar, 19 OTTs and  
30GB - all for just Rs. 200. Recharge now https://i.airtel.in/rc200s1l, creator=com.google.android.apps.messaging, seen=1  
Row: 4 _id=3355, thread_id=371, address=AD-TRAIND-G, person=NULL, date=1788646015464, date_sent=1788640147000, protocol=0, read=0, status=1, type=1, reply_path_present=0,  
subject=proto:cjokIMkVbSnb29nbgUwYskmcpZC5chHBZLml1c3NhZ2Zluz4SFClAKHc4asmbFvdCSAHT3IK9Mzz1, body=ಮುಕ್ತಿ ಏನಿದ್ದರೆ ವಸರಿಗೆ ನನ್ನನ್ನು ಸೇರಿಸಿ ಒಮ್ಮೆ ನಾನು ತ್ರಾಸಿ ನೋಡುತ್ತೇನೆ. ಶಕ ಮಗನಾಗದು  
ಅಜ್ಜನವಿದೆ ಅಂತರಿಗೆ ಕೆ ಮುಸ್ಲಿಂಗೆ ನಿಂದದ್ದಿನಿಂದೆಲ್ಲೆ,ಈ ವಿಷಯವು ನಿಂಬೂದತ್ತೆ ಮುಕ್ತಿದೆ ನೆರೆಸ್ಸು ಓದುದ್ದೆರೆ ಕರು ರಿಸ್ಸೆರೆ ದೆಯೊದುದುಂಡುಂ, service_center=+919890784018, locked=0, sub_id=1, error_code=-1,  
creator=com.google.android.apps.messaging, seen=1  
Row: 5 _id=3354, thread_id=378, address=AD-AIRDOT-S, person=NULL, date=1788672676749, date_sent=1788672668000, protocol=0, read=0, status=-1, type=1, reply_path_present=0,  
subject=proto:cjokIMkVbSnb29nbgUwYskmcpZC5chHBZLml1c3NhZ2Zluz4SFClAKHciY9AhYuVcsZR9oAXGKMA, body=Alert: On receiving any unwanted SMS , please complain by calling 1909 or send SMS to  
1909 in format 'SMS content, Sender no, dd/mm/yy' or visit bit.ly/zqbKVP or report through Airtel Thunks App., service_center=+919890784018, locked=0, sub_id=1, error_code=-1,  
creator=com.google.android.apps.messaging, seen=1  
Row: 6 _id=3353, thread_id=701, address=VO-MMPTR1-P, person=NULL, date=1788615626927, date_sent=1788615625000, protocol=0, read=0, status=-1, type=1, reply_path_present=0,  
subject=proto:cjokIMkVbSnb29nbgUwYskmcpZC5chHBZLml1c3NhZ2Zluz4SFClAKHD0A3VDYJCpe9jmCB8xc/, body=SPECIAL DEAL on flights to Bengaluru: Up to Rs.3000 OFF*, Use Code: RWYVSP. Book now -  
app.mvty.com/XmzV/Ozjfc78 | Team MakeMyTrip, service_center=+919811002444, locked=0, sub_id=1, error_code=-1, creator=com.google.android.apps.messaging, seen=1  
Row: 7 _id=3352, thread_id=517, address=AD-6500024-P, person=NULL, date=1788511060005, date_sent=1788510930000, protocol=0, read=0, status=-1, type=1, reply_path_present=0,  
subject=proto:cjokIMkVbSnb29nbgUwYskmcpZC5chHBZLml1c3NhZ2Zluz4SFClAKHdipX9AwBAKFEr1X0Ldc4, body=Did you know you can get an Airtel Prepaid SIM online & get it delivered at home at no  
extra cost? Get yours now https://i.airtel.in/gKNQND0v5b, service_center=+919804560803, locked=0, sub_id=1, error_code=-1, creator=com.google.android.apps.messaging, seen=1  
Row: 8 _id=3351, thread_id=700, address=JM-MMPTR1-S, person=NULL, date=1788604799045, date_sent=1788604798000, protocol=0, read=0, status=-1, type=1, reply_path_present=0,  
subject=proto:cjokIMkVbSnb29nbgUwYskmcpZC5chHBZLml1c3NhZ2Zluz4SFClAKHciPye2URZU7IOGkyNfME, body=[#] 991058 is the OTP to create your MakeMyTrip account. We don't ask for your OTP/bank  
info. Don't share it with anyone. FKkn01u5MA - MWT, service_center=+91702195071, locked=0, sub_id=1, error_code=-1, creator=com.google.android.apps.messaging, seen=1  
Row: 9 _id=3350, thread_id=573, address=AD-AAHADR-T, person=NULL, date=1788596630950, date_sent=1788596629000, protocol=0, read=1, status=-1, type=1, reply_path_present=0,  
subject=proto:cjokIMkVbSnb29nbgUwYskmcpZC5chHBZLml1c3NhZ2Zluz4SFClAKHciVx1ql9Ab0YS1jhdw2VB, body=141419 is OTP for Aadhaar (XX9435) (valid for 10 mins) at NHA. Mandatory Biometric  
update for age group 5 to 17 is free of cost -UIDAI, service_center=+919845060672, locked=0, sub_id=1, error_code=-1, creator=com.google.android.apps.messaging, seen=1  
Row: 10 _id=3349, thread_id=461, address=AA-AIRTEL-P, person=NULL, date=1788510388284, date_sent=1788510385000, protocol=0, read=1, status=-1, type=1, reply_path_present=0,  
subject=proto:cjokIMkVbSnb29nbgUwYskmcpZC5chHBZLml1c3NhZ2Zluz4SFClAKHdiPSwDRBCGZ3SKMKxtXtc, body=Introducing Airtel2's Box Office Pack! Get Amazon Prime, Zee5, JioHotstar, 19 OTTs and  
30GB - all for just Rs. 200. Recharge now https://i.airtel.in/rc200s1l, creator=com.google.android.apps.messaging, seen=1  
Row: 11 _id=3348, thread_id=380, address=AD-AIRTEL-S, person=NULL, date=1788590867979, date_sent=1788590864000, protocol=0, read=1, status=-1, type=1, reply_path_present=0,  
subject=proto:cjokIMkVbSnb29nbgUwYskmcpZC5chHBZLml1c3NhZ2Zluz4SFClAKHbiyAFROLtITotWLD9, body=Never respond to emails/instantly added links in messages/calls asking you to update or verify  
KYC documents/User ID/Password/Debit Card Number/PIN/CVV/OTP etc. Immediately change your passwords/CVV/PIN if you have accidentally revealed your credentials., service_center=+  
919831029607, locked=0, sub_id=1, error_code=-1, creator=com.google.android.apps.messaging, seen=1  
Row: 12 _id=3347, thread_id=461, address=AA-AIRTEL-P, person=NULL, date=1788428428528, date_sent=1788428479000, protocol=0, read=1, status=-1, type=1, reply_path_present=0,  
subject=proto:cjokIMkVbSnb29nbgUwYskmcpZC5chHBZLml1c3NhZ2Zluz4SFClAKHGmi/yfjtFLtSHImrbT3f, body>You're now an Airtel Insider: Enjoy FREE Pocket FM Premium & EazyDiner Prime. Claim now
```

5. Output of Extracted Call Logs (call_logs.txt)

```

1: 0 formatted-number=NULL, duration=0, subject=NULL, is call log phone account migration pending=0, encrypt=0, is smart answer=0, subscription_id=1, photo_id=0, post_dial_digits=
2: call screening app name=NULL, priority=0, number=+917396283848, countryiso-IN, smart answer data=NULL, recent=1, smart answer record id=1, photo uri=NULL, geocoded location=India,
3: vtcall=1, missed reason=0, is business call=0, block reason=0, subscription component name=com.android.phone/com.android.services.telephony.TelephonyConnectionService, is_secret=0,
4: add for all users=0, numbertype=2, conf_callid=NULL, attribution=NULL, features=64, transcription=NULL, record duration=NULL, last modified=1784303423413, id=11194, new=0, cnap=
5: Sudesh
6: Anna, date=1761754437657, name=Sudesh Anna, type=2, simid=1, is_child=0, is_multi=0, call category=0, carrier=1, smart record data=NULL, presentation=1, via number=, channel=0,
7: numberlabel=NULL, normalized number=+917396283848, composer photo uri=NULL, phone account address=, phone account hidden=0, lookup uri=NULL, record path=NULL, voicemail uri=NULL,
8: matched number=NULL, smart translate data=NULL, extension label=NULL, carrier extension label=NULL, transcription state=0, data usage=NULL, ai anti fraud data=NULL, location=NULL,
9: asserted display name=NULL, call screening component name=NULL, number mark label=254, is read=1
10: Row: 1 formatted-number=NULL, duration=12, subject=NULL, is call log phone account migration pending=0, encrypt=0, is smart answer=0, subscription_id=1, photo_id=0, post_dial_digits=
11: call screening app name=NULL, priority=0, number=+917396283848, countryiso-IN, smart answer data=NULL, recent=1, smart answer record id=1, photo uri=NULL, geocoded location=India,
12: vtcall=1, missed reason=0, is business call=0, block reason=0, subscription component name=com.android.phone/com.android.services.telephony.TelephonyConnectionService, is_secret=0,
13: add for all users=0, numbertype=2, conf_callid=NULL, attribution=NULL, features=64, transcription=NULL, record duration=NULL, last modified=1784303423413, id=11195, new=0, cnap=
14: +917396283848, date=1761754634845, name=Sudesh Anna, type=1, simid=1, is_child=0, is_multi=0, call category=0, carrier=1, smart record data=NULL, presentation=1, via number=, channel=0,
15: numberlabel=NULL, normalized number=+917396283848, composer photo uri=NULL, phone account address=, phone account hidden=0, lookup uri=NULL, record path=NULL, voicemail uri=NULL,
16: matched number=NULL, smart translate data=NULL, extension label=NULL, carrier extension label=NULL, transcription state=0, data usage=NULL, ai anti fraud data=NULL, location=NULL,
17: asserted display name=NULL, call screening component name=NULL, number mark label=254, is read=1
18: Row: 2 formatted-number=NULL, duration=9, subject=NULL, is call log phone account migration pending=0, encrypt=0, is smart answer=0, subscription_id=1, photo_id=0, post_dial_digits=
19: call screening app name=NULL, priority=0, number=+917396283848, countryiso-IN, smart answer data=NULL, recent=1, smart answer record id=1, photo uri=NULL, geocoded location=India,
20: vtcall=1, missed reason=0, is business call=0, block reason=0, subscription component name=com.android.phone/com.android.services.telephony.TelephonyConnectionService, is_secret=0,
21: add for all users=0, numbertype=2, conf_callid=NULL, attribution=NULL, features=64, transcription=NULL, record duration=NULL, last modified=1784303423413, id=11196, new=0, cnap=
22: +917396283848, date=1761754835010, name=Sudesh Anna, type=1, simid=1, is_child=0, is_multi=0, call category=0, carrier=1, smart record data=NULL, presentation=1, via number=, channel=0,
23: numberlabel=NULL, normalized number=+917396283848, composer photo uri=NULL, phone account address=, phone account hidden=0, lookup uri=NULL, record path=NULL, voicemail uri=NULL,
24: matched number=NULL, smart translate data=NULL, extension label=NULL, carrier extension label=NULL, transcription state=0, data usage=NULL, ai anti fraud data=NULL, location=NULL,
25: asserted display name=NULL, call screening component name=NULL, number mark label=254, is read=1
26: Row: 3 formatted-number=NULL, duration=194, subject=NULL, is call log phone account migration pending=0, encrypt=0, is smart answer=0, subscription_id=1, photo_id=0, post_dial_digits=
27: call screening app name=NULL, priority=0, number=+917396283848, countryiso-IN, smart answer data=NULL, recent=1, smart answer record id=1, photo uri=NULL, geocoded location=India,
28: vtcall=1, missed reason=0, is business call=0, block reason=0, subscription component name=com.android.phone/com.android.services.telephony.TelephonyConnectionService, is_secret=0,
29: add for all users=0, numbertype=2, conf_callid=NULL, attribution=NULL, features=64, transcription=NULL, record duration=NULL, last modified=1784303423413, id=11197, new=0, cnap=
30: +917396283848, date=1761754854622, name=Sudesh Anna, type=1, simid=1, is_child=0, is_multi=0, call category=0, carrier=1, smart record data=NULL, presentation=1, via number=, channel=0,
31: numberlabel=NULL, normalized number=+917396283848, composer photo uri=NULL, phone account address=, phone account hidden=0, lookup uri=NULL, record path=NULL, voicemail uri=NULL,
32: matched number=NULL, smart translate data=NULL, extension label=NULL, carrier extension label=NULL, transcription state=0, data usage=NULL, ai anti fraud data=NULL, location=NULL,
33: asserted display name=NULL, call screening component name=NULL, number mark label=254, is read=1
34: Row: 4 formatted-number=NULL, duration=7, subject=NULL, is call log phone account migration pending=0, encrypt=0, is smart answer=0, subscription_id=1, photo_id=0, post_dial_digits=
35: call screening app name=NULL, priority=0, number=+91939229395, countryiso-IN, smart answer data=NULL, recent=1, smart answer record id=1, photo uri=NULL, geocoded location=India,
36: vtcall=1, missed reason=0, is business call=0, block reason=0, subscription component name=com.android.phone/com.android.services.telephony.TelephonyConnectionService, is_secret=0,
37: add for all users=0, numbertype=2, conf_callid=NULL, attribution=NULL, features=64, transcription=NULL, record duration=NULL, last modified=1784303423413, id=11198, new=0, cnap=
38: +91939229395, date=1761755212323, name=Sairam S16, type=3, simid=1, is_child=0, is_multi=0, call category=0, carrier=1, smart record data=NULL, presentation=1, via number=, channel=0,
39: numberlabel=NULL, normalized number=+91939229395, composer photo uri=NULL, phone account address=, phone account hidden=0, lookup uri=NULL, record path=NULL, voicemail uri=NULL,
40: matched number=NULL, smart translate data=NULL, extension label=NULL, carrier extension label=NULL, transcription state=0, data usage=NULL, ai anti fraud data=NULL, location=NULL,
41: asserted display name=NULL, call screening component name=NULL, number mark label=254, is read=1
42: Row: 5 formatted-number=NULL, duration=10, subject=NULL, is call log phone account migration pending=0, encrypt=0, is smart answer=0, subscription_id=1, photo_id=0, post_dial_digits=
43: call screening app name=NULL, priority=0, number=+91939229395, countryiso-IN, smart answer data=NULL, recent=1, smart answer record id=1, photo uri=NULL, geocoded location=India,
44: vtcall=1, missed reason=0, is business call=0, block reason=0, subscription component name=com.android.phone/com.android.services.telephony.TelephonyConnectionService, is_secret=0,

```

6. Extracted Device Properties (Model, Android Version, Security Patch Level)

```
PS C:\Users\saiha\Downloads\platform-tools-latest-windows\platform-tools> notepad .\sms_extraction.txt
PS C:\Users\saiha\Downloads\platform-tools-latest-windows\platform-tools> notepad .\call_logs.txt
PS C:\Users\saiha\Downloads\platform-tools-latest-windows\platform-tools> Write-Host "Device Model:" (.\adb.exe shell getprop ro.product.model)
Device Model: I2302
PS C:\Users\saiha\Downloads\platform-tools-latest-windows\platform-tools> Write-Host "Android Version:" (.\adb.exe shell getprop ro.build.version.release)
Android Version: 16
PS C:\Users\saiha\Downloads\platform-tools-latest-windows\platform-tools> Write-Host "Security Patch Level:" (.\adb.exe shell getprop ro.build.version.security_patch)
```

7. Uninstallation of AFLogical OSE from the Device

```
PS C:\Users\saiha\Downloads\platform-tools-latest-windows\platform-tools> .\adb.exe uninstall --user 0 com.viaforensics.android.aflogical_ose
Success
PS C:\Users\saiha\Downloads\platform-tools-latest-windows\platform-tools> |
```

Rubrics

Criteria & Marks Assigned	Mark Allotted	Mark Awarded
1. GitHub Activity & Submission Regularity	3	
2. Application of Forensic Tools & Practical Execution	3	
3. Documentation & Reporting	2	
4. Engagement, Problem-Solving & Team Collaboration	2	
Total	10	

Result: Successfully acquired and analyzed volatile memory from an Android environment using LiME, and investigated memory artifacts using Volatility 3.